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Adaptive Identity Security Framework for Agentic AI Systems: Architecture, Implementation, and Performance Evaluation

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ABSTRACT: The increasing autonomy of agentic artificial intelligence (AI) systems introduces a fundamental challenge for identity security because conventional identity and access mechanisms are primarily designed around relatively stable human users, applications, and service accounts. Agentic AI systems can dynamically select tools, invoke services, exchange information, and perform multi-step actions with limited human intervention. Consequently, identity security must evolve from static authentication toward adaptive mechanisms capable of continuously evaluating identity, context, authorization, behavioral patterns, and operational risk. This paper proposes an Adaptive Identity Security Framework for Agentic AI Systems that integrates identity representation, contextual authentication, dynamic authorization, behavioral monitoring, risk-based policy evaluation, and continuous identity assurance. The framework is conceptually positioned around an identity-fabric approach in which agent identities remain traceable across heterogeneous services and dynamically changing execution contexts. The methodology defines architectural layers, an adaptive authorization model, an identity-risk evaluation process, and an implementation workflow. The supplied literature is also critically examined to identify transferable methodological insights concerning artificial intelligence, automated decision systems, model reliability, and evaluation. The analysis indicates that adaptive identity security can reduce excessive privilege, improve action traceability, and provide stronger controls over autonomous AI operations, although it introduces computational overhead, policy complexity, identity-provenance challenges, and potential false-positive authorization decisions. The framework extends the identity-fabric perspective proposed by Pappu, Bhushan, and Jaiswal by emphasizing continuous adaptation and performance evaluation as core properties of agentic AI identity management.

KEYWORDS: Agentic AI, Identity Security, Adaptive Authorization, Identity Fabric, Zero-Trust Architecture, AI Agents, Behavioral Risk, Continuous Authentication, Dynamic Access Control, AI Security.

1. Introduction

Agentic AI represents a transition from conventional predictive or conversational AI toward systems capable of pursuing objectives through sequential decisions, tool invocation, service interaction, and autonomous task execution. This evolution changes the security meaning of identity. A conventional application may operate under a relatively stable service identity, whereas an AI agent can dynamically initiate actions across APIs, databases, cloud services, enterprise applications, and external tools. The resulting security problem is therefore not simply determining whether an entity is authenticated; it is determining whether a particular agent should be permitted to perform a particular action under a particular context.

The challenge becomes more significant when an agent can generate intermediate objectives and select tools independently. A valid agent identity does not automatically imply that every action generated by that agent should be authorized. Identity security must consequently distinguish between who or what the agent is, what it is currently attempting to do, which resources it is accessing, and whether its behavior remains consistent with established policy. Pappu, Bhushan, and Jaiswal's identity-fabric perspective provides an important conceptual basis for future-proofing identity security for agentic AI by treating identity as an integrated security capability rather than an isolated authentication mechanism (Pappu, Bhushan, and Jaiswal, 2026).

The research problem addressed in this paper is the absence of a sufficiently adaptive identity architecture for autonomous AI environments in which identities, privileges, execution contexts, and actions may change

dynamically. Static role-based access control can be inadequate when an agent's risk changes during execution. Similarly, authentication performed only at session initiation cannot adequately address continuously changing behavior.

The objective of this research is to develop a conceptual adaptive identity security framework that combines persistent agent identity, contextual risk evaluation, dynamic authorization, behavioral monitoring, and continuous policy enforcement. The framework is designed for enterprise AI environments in which agents interact with multiple services and where accountability for autonomous decisions is required.

The scope includes architectural design, authorization logic, identity-risk evaluation, implementation considerations, and performance evaluation. The study does not claim empirical validation using a production deployment. Instead, it develops a research-driven framework and identifies measurable performance dimensions for subsequent implementation.

2. Literature Review

The supplied literature is heterogeneous and is predominantly concerned with artificial intelligence, automated diagnosis, machine learning, and deep-learning systems rather than identity security. This distinction is important because directly treating these publications as cybersecurity evidence would create an inappropriate literature synthesis. Nevertheless, their methodological emphasis on automation, model evaluation, system reliability, and AI-assisted decision-making provides transferable conceptual insights for constructing an adaptive security framework.

Birdsong's work on automated cervical cytology screening represents an early example of computational automation applied to a traditionally expert-driven process (Birdsong, 1996). Its relevance to the present research is methodological rather than domain-specific: when computational systems participate in decisions previously dependent on human judgment, mechanisms for controlling, evaluating, and interpreting automated outputs become increasingly important.

Karampotsis, Dounias, and Jantzen examine machine-learning approaches for Pap-smear diagnosis and present the broader evolution of intelligent analytical techniques (Karampotsis, Dounias, and Jantzen, 2019). This evolution illustrates how AI systems can move from predefined analytical procedures toward increasingly sophisticated computational decision processes. For agentic AI, the analogous transition creates a security requirement: as decision autonomy increases, identity controls must become capable of evaluating not only system membership but also action legitimacy.

Jusman, Ng, and Abu Osman discuss intelligent screening systems for cervical cancer, demonstrating the role of computational intelligence in supporting automated screening processes (Jusman, Ng, and Abu Osman, 2014). The broader methodological lesson is that intelligent systems operate within a decision environment where input, processing, and output must be connected to an accountable system architecture.

Chitra and Kumar provide a review of automated cervical-cancer screening developments, highlighting the progression of intelligent technologies for diagnostic support (Chitra and Kumar, 2021). Similarly, Mustafa et al. analyze cervical-cancer detection techniques chronologically, demonstrating the continuous evolution of computational approaches (Mustafa et al., 2023). These studies support the proposition that security architectures for AI should be designed as evolving systems rather than fixed controls.

Rahaman's survey of deep-learning approaches for cervical cytopathology image analysis and Srinidhi, Ciga, and Martel's survey of deep neural networks in computational histopathology further demonstrate the increasing complexity of AI models and computational pipelines (Rahaman, 2020; Srinidhi, Ciga, and Martel, 2021). As AI architectures become more complex, their surrounding operational controls must likewise address provenance, execution integrity, and accountability.

Khare et al. explicitly focus on systematic evaluation and research recommendations surrounding AI-based cervical-cancer detection (Khare et al., 2024). This emphasis on systematic evaluation is particularly relevant to the present framework because identity security for agentic AI cannot be assessed solely through successful authentication rates. Evaluation must include authorization accuracy, policy response time, privilege reduction, detection capability, and operational overhead.

Youneszade, Marjani, and Pei discuss deep learning in cervical-cancer diagnosis while identifying opportunities and open research challenges (Youneszade, Marjani, and Pei, 2023). The identification of open challenges reinforces the importance of designing AI-related architectures that remain adaptable as technologies and operational requirements change.

The WHO references establish the broader importance of structured health information and cancer-related systems (WHO, 2022a; WHO, 2022b). Although they do not directly address agentic AI security, they reinforce the importance of reliable information handling in sensitive computational environments.

The principal research gap is therefore clear: the supplied literature demonstrates extensive development of intelligent computational systems but does not directly establish an adaptive identity-security architecture for autonomous AI agents. Pappu, Bhushan, and Jaiswal directly address the identity-security problem in agentic AI through an identity-fabric perspective, making their work the primary conceptual anchor for the proposed framework (Pappu, Bhushan, and Jaiswal, 2026). The present study extends that perspective by emphasizing continuous adaptation, risk-based authorization, behavioral monitoring, and measurable performance evaluation.

3. Methodology

3.1 Research Design

This study adopts a conceptual framework-development methodology. Rather than training an AI model or evaluating a commercial identity platform, the research decomposes the identity problem of agentic AI into functional security components and integrates them into an adaptive architecture.

The proposed design assumes that an AI agent possesses a persistent machine-readable identity. That identity is associated with attributes including agent type, owner or controlling entity, permitted capabilities, operational environment, credential status, trust level, and historical behavior. The architecture then evaluates each significant action against contextual security policies.

The methodology follows five principal stages: identity establishment, context collection, risk evaluation, dynamic authorization, and continuous monitoring.

3.2 Adaptive Identity Architecture

The proposed architecture consists of six logical layers.

Identity Layer.

This layer creates persistent identities for AI agents, tools, services, and human supervisors. Instead of relying on a generic service account, every autonomous agent receives an independently traceable identity. Identity metadata includes ownership, purpose, permitted functions, credential state, and lifecycle status.

Authentication and Assurance Layer.

Authentication establishes the initial legitimacy of the agent. However, the proposed framework treats authentication as a continuous assurance process rather than a single event. Changes in execution environment, credential condition, behavior, or task sensitivity can trigger additional identity verification.

Context Layer.

The context engine collects information about the requested resource, action type, data sensitivity, execution environment, time, agent history, and relationship between the agent and requested capability. This enables the framework to distinguish low-risk actions from high-impact operations.

Risk Evaluation Layer.

A conceptual risk score can be represented as:

[

$$R = w_I I + w_C C + w_B B + w_A A + w_D D$$

]

where (I) represents identity assurance, (C) represents contextual risk, (B) represents behavioral deviation, (A) represents action sensitivity, and (D) represents data sensitivity. The weights (w) can be calibrated according to organizational requirements.

A low-risk score may permit the action automatically. A moderate score may require additional verification or reduced privileges, while a high score may cause the action to be denied or escalated to human approval.

Authorization Layer.

Authorization is performed at action level rather than merely session level. For example, an AI agent may be authorized to retrieve customer records but not export an entire database. This distinction is essential because an authenticated identity can still attempt an unauthorized operation.

Monitoring and Audit Layer.

Every significant agent action is recorded with identity, timestamp, requested capability, policy decision, contextual risk, resource, and outcome. This creates a traceable identity fabric across the agent's lifecycle.

3.3 Dynamic Authorization Model

The framework adopts a continuous decision process:

[

$$\text{Decision} = f(\text{Identity}, \text{Context}, \text{Risk}, \text{Policy}, \text{Behavior})$$

]

An authorization decision is therefore not permanent. Suppose an enterprise assistant is initially authorized to read internal documents. If the same agent subsequently attempts to access sensitive financial records after an unusual privilege escalation, its behavioral and

contextual risk values increase. The policy engine can respond by restricting the operation even though the agent remains authenticated.

This model reduces the weakness of static authorization. It also aligns with the broader principle that increasingly autonomous AI systems require identity architectures capable of adapting to changing operational conditions (Pappu, Bhushan, and Jaiswal, 2026).

3.4 Identity Lifecycle Management

Agent identity should follow a lifecycle consisting of provisioning, activation, monitoring, modification, suspension, and retirement. New agents receive the minimum capabilities necessary for their assigned purpose. Capability changes should generate auditable policy events rather than silently modifying privileges.

A hypothetical enterprise deployment illustrates this approach. An AI procurement agent may be permitted to search suppliers and prepare purchase recommendations. It may not initially possess authority to execute payments. If a business process requires payment execution, a separate authorization policy can temporarily grant narrowly scoped permission subject to contextual verification and approval.

3.5 Performance Evaluation Method

The proposed framework can be evaluated using five principal indicators.

First, authorization accuracy measures whether legitimate actions are permitted and unauthorized actions are rejected. Second, policy decision latency measures the time required to produce an authorization decision. Third, privilege exposure measures the amount of unnecessary access granted to agents. Fourth, behavioral detection effectiveness measures the ability to identify deviations from established activity patterns. Fifth, audit completeness measures whether agent actions can be reconstructed from identity and policy records.

A composite performance score can be represented as:

[

P = \alpha A_c + \beta D_e + \gamma T_a + \delta L_c - \epsilon O

]

where (A_c) is authorization accuracy, (D_e) is detection effectiveness, (T_a) is audit completeness, (L_c) represents latency performance, and (O) represents operational overhead.

The framework therefore treats performance as a multidimensional security problem rather than reducing evaluation to authentication speed alone.

4. Results

The conceptual analysis produces several principal findings concerning the requirements of identity security in agentic AI environments.

First, persistent identity is necessary but insufficient. An agent can be strongly authenticated and still perform an inappropriate action. The proposed framework therefore separates identity assurance from authorization. This distinction is particularly important for autonomous systems because the same identity may perform multiple actions with substantially different security consequences. The identity-fabric approach provides a suitable foundation for maintaining continuity of identity across these interactions (Pappu, Bhushan, and Jaiswal, 2026).

Second, dynamic authorization provides stronger security than static privilege assignment for environments in which agent behavior changes during execution. A static role may provide broad capabilities for convenience, but those capabilities can become dangerous when an agent is compromised, manipulated, or operating outside its expected context. Risk-based authorization enables privileges to change without requiring the entire identity to be invalidated.

Third, behavioral context becomes a security signal. Agentic systems produce activity patterns that can be evaluated against historical or policy-defined behavior. An unexpected sequence of actions can therefore contribute to a risk escalation. This concept complements the systematic emphasis on evaluation found in AI literature, where the reliability of automated systems must be examined rather than assumed (Khare et al., 2024).

Fourth, least privilege becomes dynamic rather than static. The proposed architecture grants capabilities according to current task requirements and contextual conditions. This reduces privilege exposure because an agent does not need to retain every capability throughout its lifecycle.

Fifth, traceability is a core architectural property. Every agent-generated operation should be associated with an identifiable principal, requested capability, policy decision, and execution outcome. Without this chain, organizations may be unable to determine which autonomous component initiated a sensitive action.

Sixth, the framework introduces measurable trade-offs. Continuous risk assessment improves security visibility but increases computational and policy-processing overhead. More restrictive authorization can reduce unauthorized activity but may also generate false

positives and interrupt legitimate workflows. Therefore, performance evaluation must consider both security effectiveness and operational efficiency.

Finally, the analysis confirms a substantial literature gap. The supplied AI literature demonstrates significant progress in automated computational decision systems (Birdsong, 1996; Rahaman, 2020; Srinidhi, Ciga, and Martel, 2021), while direct treatment of identity security for autonomous AI is comparatively specialized. This gap supports further research into empirically validated identity-fabric architectures for agentic systems.

5. Discussion

The proposed framework has both theoretical and practical implications. Theoretically, it reframes identity security from an authentication-centered concept into a continuous decision architecture. In conventional systems, identity often functions as a relatively stable input to authorization. Agentic AI challenges this assumption because autonomous behavior introduces changing execution contexts, dynamically selected tools, and potentially unpredictable action sequences. Consequently, identity must remain persistent while authorization becomes adaptive.

The framework extends the identity-fabric perspective by positioning the agent identity as a security anchor across the complete action lifecycle. Pappu, Bhushan, and Jaiswal emphasize future-proof identity security for agentic AI through an identity-fabric perspective; the present framework builds on this concept by explicitly connecting identity with behavioral risk and continuous policy evaluation (Pappu, Bhushan, and Jaiswal, 2026).

The literature concerning intelligent automated systems also highlights an important parallel. AI systems have progressively evolved from conventional computational approaches toward deep-learning and increasingly complex automated decision processes (Karampotsis, Dounias, and Jantzen, 2019; Youneszade, Marjani, and Pei, 2023). Security architecture must evolve correspondingly. A security model designed for predictable software processes may not be sufficient for systems that dynamically select actions.

A major practical implication is the need to separate identity, capability, and authorization. An enterprise agent should not receive unrestricted access simply because it has been authenticated. Instead, capabilities should be explicitly scoped and dynamically evaluated. This approach can reduce blast radius when credentials are compromised or when an agent behaves unexpectedly.

However, adaptive authorization also introduces contradictions. A highly sensitive policy may improve protection but reduce usability. Frequent verification may increase latency and disrupt autonomous workflows. Similarly, behavioral monitoring can generate false positives when legitimate agent behavior differs from historical patterns. The optimal framework must therefore balance security sensitivity against operational continuity.

Another limitation concerns the quality of contextual information. Risk-based authorization is only as effective as the signals supplied to the decision engine. Missing, stale, or manipulated context can lead to inappropriate decisions. Agent identity provenance is also challenging in multi-agent environments where one agent delegates tasks to another. The architecture must preserve a chain of responsibility so that downstream actions remain attributable to their initiating principal.

The principal methodological limitation of this study is that the proposed framework is conceptual rather than empirically validated. No production dataset or controlled enterprise deployment was supplied. Consequently, the performance equations represent evaluation mechanisms rather than measured outcomes. Future empirical research should implement the architecture in a controlled multi-agent environment and compare static and adaptive authorization under identical workloads.

The cross-domain nature of the supplied literature is another limitation. Most provided studies concern AI-assisted cervical-cancer screening rather than cybersecurity. Their contribution to this study is therefore methodological and conceptual rather than direct evidence for identity-security controls. This limitation should be explicitly recognized in any empirical extension of the framework.

6. Conclusion

This paper proposed an Adaptive Identity Security Framework for Agentic AI Systems that integrates persistent agent identity, continuous assurance, contextual risk evaluation, dynamic authorization, behavioral monitoring, and comprehensive auditing. The central argument is that authentication alone is inadequate for autonomous AI because a legitimate identity may still perform an inappropriate action.

The framework contributes a structured architecture in which identity functions as a persistent security anchor while authorization continuously adapts to context, behavior, resource sensitivity, and action risk. This model supports least privilege, improves traceability, and provides a mechanism for responding to changing operational conditions.

The analysis also identified a clear research gap between the broader literature on intelligent automated

systems and the emerging problem of identity security for autonomous AI agents. The supplied AI literature demonstrates the evolution and increasing complexity of automated decision systems, while the identity-fabric perspective of Pappu, Bhushan, and Jaiswal provides the most direct conceptual foundation for agentic AI identity security (Pappu, Bhushan, and Jaiswal, 2026).

Future research should implement the framework in realistic multi-agent environments, evaluate policy-decision latency and authorization accuracy, test adversarial identity scenarios, and investigate federated identity across heterogeneous enterprise platforms. Particular attention should be given to agent-to-agent delegation, identity provenance, adaptive privilege reduction, behavioral anomaly detection, and human approval mechanisms for high-impact operations. Such empirical validation would determine whether adaptive identity security can provide measurable improvements over conventional static access-control approaches while maintaining acceptable operational performance.

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